

ORGAN SYSTEM ANATOMY (MID & POST LEC)

Lec #3: Embryology & Brodmann's & Circle of Willis | OPHT 1-1 | PRELIMS

By Accel Jimenez

GERM LAYERS

- ★ **Ectoderm**
 - Skin, hair
 - Nervous System
- ★ **Mesoderm**
 - Structural framework of the body
 - Muscle, skeleton, kidneys, reproductive system.
- ★ **Endoderm**
 - Internal organs
 - Lining of endocrine glands, lungs, digestive tract, liver, stomach.

* trilaminar disc is formed during the **third week** of embryonic development

NEURULATION

- Induction begins with the notochord, which secretes signaling molecules that direct ectodermal cells to become neuroectoderm.
- Neural plate: thickened ectodermal cells on the dorsal surface.
- Neural plate folds and fuses - neural tube (day 22-23)
 - **Neural tube** - forms CNS (brain + spinal cord)
 - **Neural Crest cells** - forms PNS + other structures.

*Neurulation means the formation of the nervous system.

NEURULATION PROCESS

- Induction begins with the **notochord**, which secretes signaling molecules that direct ectodermal cells to become neuroectoderm.
- **Neural plate**: thickened ectodermal cells on the dorsal surface.
- Neural groove and neural folds form by invagination.
- Neural tube forms when folds fuse (**day 22-23**).
 - **Cranial end** → brain.
 - **Caudal end** → spinal cord.
- Anterior neuropore closes (**~day 25**) → future brain.
- Posterior neuropore closes (**~day 27**) → future spinal cord.

- Neural crest cells migrate laterally → form PNS structures (dorsal root ganglia, Schwann cells, autonomic ganglia, melanocytes, etc.).

CNS PATTERNING AXIS

1. **Anterior-Posterior Axis (Head-to-tail)**
 - **Cranial end** - develops into 3 primary vesicles.
 - Forebrain (Prosencephalon)
 - Midbrain (Mesencephalon)
 - Hindbrain (Rhombencephalon)
 - **Caudal end** - develops into the spinal cord.

2. **Dorsal Ventral Axis (Back-to-Belly)**
Ventral = anterior, Dorsal = Back

- Defined by Chemical signaling molecules
- **Ventralizing signal**:
 - Matures into **motor neurons in the ventral horn** in the spinal cord.
 - **Sonic Headshot**: cells that function to **facilitate formation/maturation**
- **Dorsalizing signal**:
 - Matures into **sensory neurons in the dorsal horn** in the spinal cord.

Neural Tube Defects (reason: **folic acid deficiency, MTHR genes, infections,**)

- **Anencephaly**
 - No brain formed
 - Defect was on the cranial end
- **Spina Bifida**
 - Spinal cord did not fully develop
 - Defect on caudal end

PRIMITIVE BRAIN DEVELOPMENT

- **Prosencephalon (Forebrain)**
 - **Telencephalon**
 - Matures into **Cerebral Cortex** and **Basal Ganglia**
 - **Diencephalon**
 - Matures into **thalamus, hypothalamus, epithalamus**
- **Mesencephalon (Midbrain)**
 - Matures into **midbrain**

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- **Rhombencephalon (Hindbrain)**
 - **Metencephalon**
 - Matures into cerebellum and pons
 - **Myelencephalon**
 - Matures into medulla oblongata

GYRUS, SULCI, & FISSURES

Gyrus

- Raised **fold**s, ridge of the cerebral cortex (intestines)
- Function: increases surface area for neurons.

Sulcus

- Shallow grooves, lines in the brain between gyri
- Function: divides the brain into lobes or functional areas.

Fissure

- Deep sulcus
- Divides major lobes and hemispheres
- Longitudinal fissure (cranial line in the brain)

Major Sulci and Fissures

- **Central Sulcus (Rolandic Sulcus)**
 - separates primary motor cortex (frontal lobe) from the primary somatosensory cortex (parietal lobe)
 - Landmark between precentral gyrus (motor strip) and postcentral gyrus (sensory strip)
- **Lateral Sulcus (Sylvian Fissure)**
 - Separates frontal/parietal lobes from the temporal lobe
- **Parieto-Occipital Sulcus**
 - Separates parietal from occipital lobe.
- **Calcarine Fissure** (Occipital lobe)
 - Location of primary visual cortex (area 17)
- **Transverse fissure** - Divides the cerebrum from the cerebellum

*All fissures are sulci, but not all sulci are fissures.

*The lateral sulcus is so deep that some anatomists classify it as a fissure, hence the interchangeable names: Lateral Sulcus = Sylvian Fissure.

Major Gyri

1. **Precentral Gyrus** (Frontal Lobe) → Primary Motor Cortex (BA 4)
2. **Postcentral Gyrus** (Parietal lobe) → Primary Somatosensory Cortex (BA 3,1,2).
3. **Superior Temporal Gyrus** → Primary Auditory Cortex (BA 41,42).
4. **Cingulate Gyrus** (Limbic system) → emotion, motivation, memory.

Function of Gyri and Sulci:

- Increase surface area of the cortex (more neurons)
- Functionally separate brain regions for specialized processing.

If Cerebrum Cortex lesions, Contralateral
If nerve lesions, Ipsilateral

BRODMANN'S AREAS

Brodmann's Areas

- ★ Korbinian Brodmann
- ★ Cytoarchitecture (different cell types and densities)

Frontal Lobe:

- ★ **Area 4**
 - Primary Motor area - responsible for **motor functions of the brain**.
 - Precentral Gyrus - voluntary movements
 - Lesion: *contralateral hemiplegia (weakness)*
- ★ **Area 6**
 - Pre-motor area - responsible for **initiation, planning, coordination**.
- ★ **Area 8**
 - Frontal Eye field - responsible for conjugate movement of both eyes.
- ★ **Area 9-12**
 - Pre-frontal: responsible for **intelligence, behavior, and judgement**.
- ★ **Area 44, 45**
 - Speech motor area: Responsible for Broca's Area, dominant hemisphere → speech production

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- Lesion of Brocca's Area:
Aphasia - can't talk, but can comprehend. If can talk, either stutter, or very slow.

Parietal Lobe:

★ Area 3,1,2

- Primary somatosensory area
- Post central gyrus
- Lesion: **Contralateral hemianesthesia** (loss of sensation)

★ Area 5,7

- Somatosensory Association Cortex: responsible for other sensations or tactiles.
- Lesion: Neglect Syndrome
 - **Barragnosis**: ability to recognize the weight of things placed in the hand.
 - **Graphesthesia**: ability to recognize written objects in the body.
 - **Stereognosis**: ability to recognize the object based on the shape.
 - **Propriepception**: position sense
 - **Kinesthesia**: movement sense.

★ Area 39

- Angular Gyrus
- Lesion: **Gertsmann Syndrome**

★ Area 40

- Supramarginal Gyrus
- Responsible for motor movement
- Lesion: **Ideomotor Apraxia** - Inability to execute task without
 - Ideomotor - can do it, but when asked to, can't.
 - Ideational - when asked to comb hair, used it as a toothbrush

★ Area 43

- Primary Gustatory - responsible for taste sensation
- Cranial nerve 7: anterior 2/3

Temporal lobe:

★ Area 41

- Primary auditory cortex - handles basic sound (loudness, pitch)

○ Lesions:

- Right Area 41 - Left partial deafness
- Left Area 41 - Right partial deafness
- Whole Area 41 - cortical deafness

★ Area 42

- Secondary auditory cortex - integrates/interprets sound patterns.
- Lesion: **Auditory Agnosia** (can't distinguish sound)

★ Area 22

- *Right* - part of **auditory association area** (area 22)
- *Left* - **Wernicke's area** - responsible for comprehension.
 - Lesion of Left Area 22: can talk but no comprehension.

★ Area 20, 21, 37

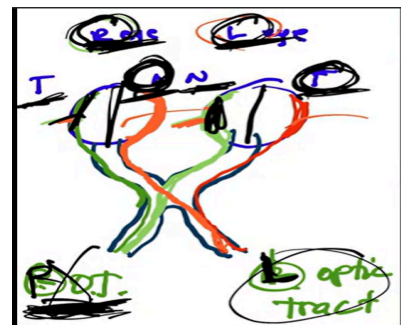
- Higher-order visual association (object recognition)

Fiber that connects **Brocca's Area** (Area 44) & **Wernicke's Area** (Area 22) is the **Arcuate Fasciculus**
Lesion: Can't talk and can't comprehend

Occipital Lobe:

● Area 17

- **Primary Visual Cortex**
- Gyrus: Striate Cortex/ Calcarine fissure.
- Lesions:
 - **Right Area 17**: Homonymous Hemianopsia
 - Vision: R nasal & L temporal side
 - Defect: R Temporal and L nasal side optic tract.



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- **Right/Left Area 17:**
Cortical Blindness (Full)
- **Area 18-19**
 - **Visual Association Area**
 - Gyrus: Parastriate Cortex
 - Lesions:
 - **18:** Dyschromatopsia (Color-blindness)
 - **18 and 19:** Visual Agnosia (can see, but can't process what it is)
 - **Prosopagnosia:** inability to recognize objects despite having normal vision.
 - **Anosagnosia:** Inability to recognize familiar faces.

CIRCLE OF WILLIS

Circle of Willis

- Originated from the **arch of aorta**.

Arch of Aorta gives rise to:

- **Brachiocephalic trunk** → Right Common Carotid + Right Subclavian
- **Left Common Carotid**
- **Left Subclavian**

Anterior Circulation:

- Common Carotid → Internal Carotid Artery (ICA) → enters skull → branches:
 - Middle Cerebral Artery (MCA) (lateral surface of brain).
 - **LARGEST ARTERY IN THE BRAIN**
 - Most commonly occluded artery in the brain
 - Largest Coverage in the brain
 - Anterior Cerebral Artery (ACA) (supplies medial frontal/parietal cortex).
 - Posterior Communicating Artery (PCOMM) (frontal to parietal area)

Posterior Circulation

- Subclavian → Vertebral Arteries → join to form Basilar Artery (largest artery in posterior circulation) → branches:

- Posterior Cerebral Arteries (PCA) (occipital lobe, visual cortex)

Anastomosis (Circle of Willis)

- Anterior Cerebral Arteries (ACA) connected by Anterior Communicating Artery (ACoA).
- Internal Carotid connects to Posterior Cerebral Artery via Posterior Communicating Artery (PCoA).
- Forms a polygonal anastomosis → collateral circulation if a vessel is blocked.

Clinical Relevance:

- Most common site of aneurysm - ACA
- Most common artery occluded - MCA
- Occlusion Patterns:
 - MCA - collateral face/arm weakness, aphasia
 - ACA - contralateral leg weakness
 - PCA - visual field defects
 - VBA - Coma/death

Posterior Cerebral Artery (PCA) is the **major artery that connects** anterior and posterior circulation.

Not everything can have lesion because of **Collateral Circulation**

HOMUNCULUS

Homunculus:

- "Little man" → cortical map of body representation in **motor (precentral gyrus)** and **sensory (postcentral gyrus)** cortices.
- **Arrangement:**
 - Head/Face - lateral
 - Upper limb/trunk - middle
 - Lower limb - medial (paracentral lobule).
- Motor Homunculus (Precentral Gyrus) - controls voluntary movement
- Sensory Homunculus (Postcentral Gyrus) - processes sensory input
- Key Concept: Body parts requiring fine control (face, hands) occupy larger cortical areas than gross-movement regions (trunk).

AREAS OF THE BRAIN

(Blood supply in relation to homunculus)

- **Anteromedial:** ACA (Lower Extremities, Genitals)
- **Lateral:** MCA (Face, Upper Extremities)
- **Posterior:** PCA (Vision, Vital organs)